Public Finance

財政學助教課

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本次助教課重點

- 居住服務的決策
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- 租稅與人力資本累積
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居住服務的決策

- 另一個重要的資本形式是自有房屋
- $R_{net} = R - I - T - D + \Delta V$

- R為持有自有房屋的利益
- I為每年為了持有自有房屋而支出的利息
- T為每年為了持有自有房屋而支出的稅費
- D為每年該自有房屋的折舊費用
- ΔV為該年度年末時，房屋的價值變動

居住服務的決策

- $R_{net} = R - I - T - D + \Delta V$
- 若該房屋出租，則持有人須負擔稅費
- 若該房屋為自用，則設算租金 (imputed rent) 不被課稅
- 更進一步，房貸利息支出可以扣除

- 結果：鼓勵買房自住

居住服務的決策

- 對自有住宅補助的批評：

- 補助有顯著的外部利益嗎？
- 補助對於所得分配有幫助嗎？

- 對自有房屋設算租金課稅不被政治人物支持

居住服務的決策

- 對自有住宅補助政策的改革建言：

- 讓房貸利息支出扣除額逐年退場
- 設定扣除額的上限
- 將扣除額 (deduction) 改成租稅扣抵 (credit)

資產組合

- Tobin：資產組合取決於預期報酬與風險程度

- 投資是風險與報酬的權衡行為
- 給定風險程度，選擇會讓報酬極大化的資產組合
- 給定報酬水準，選擇會讓風險極小化的資產組合
- 課稅之後：
- 預期報酬下降→投資減少
- 若虧損，可以從所得稅稅基中扣除→風險減少→投資增加

租稅與人力資本累積

- 簡單的投資人力資本模型：
- $B - C > 0$
- $(1 - t)B - (1 - t)C = (1 - t)(B - C) > 0$

- 假設所得效果凌駕於替代效果
- 因為租稅誘發勞工多勞動，其他條件不變下，租稅使人力資本投資更具吸引力。因此租稅會使人力資本投資增加。

租稅與人力資本累積

- 簡單投資人力資本模型的缺失：
- 人力資本投資的報酬不確定
- 忽略除了放棄收入之外的成本，例如大學學費
- 租稅制度的其他層面可以影響人力資本累積。例如對實物投資的報酬課稅
- 在累進稅制之下，B和C可能被課徵不同稅率
- 結論：理論上，對人力資本累積的報酬課稅的效果很模糊，需要有相關實證研究支持。

作業四Rosen 16.8 (d)

(Rosen 16.8) Indicate whether each of the following statements is true, false, or uncertain, and explain why:

(d) Tom's workplace provides free access to a fitness room; Jerry's does not. Horizontal equity requires that Tom be taxed on the value of having access to the fitness room.

傳統定義的水平公平要求對可以使用健身房的價值課稅；

以效用定義的水平公平不要求對可以使用健身房的價值課稅

作業四Rosen 16.8 (d)

One notion of horizontal equity is that people in equal positions should be treated equally by the tax system. Under this traditional notion of horizontal equity, the fact that Tom's workplace provides free access to a fitness room suggests this kind of compensation should be taxed; Jerry pays "full taxes" on his compensation while Tom does not. Another notion of horizontal equity relies on the utility definition of horizontal equity. This concept says that if two individuals have the same utility without taxes, they should have the same utility with taxes, and the taxes should not affect the utility ordering. One implication of the utility definition is that any existing tax structure does not violate the notion of horizontal equity if individuals are free to choose their activities and expenditures. If Tom and Jerry have free choice between the two different jobs (and identical preferences), then the net after-tax rewards (including amenities) must be the same at both jobs; otherwise there would be migration. In this case, the before-tax wage on Tom's job adjusts for the fact that there is a fringe benefit.

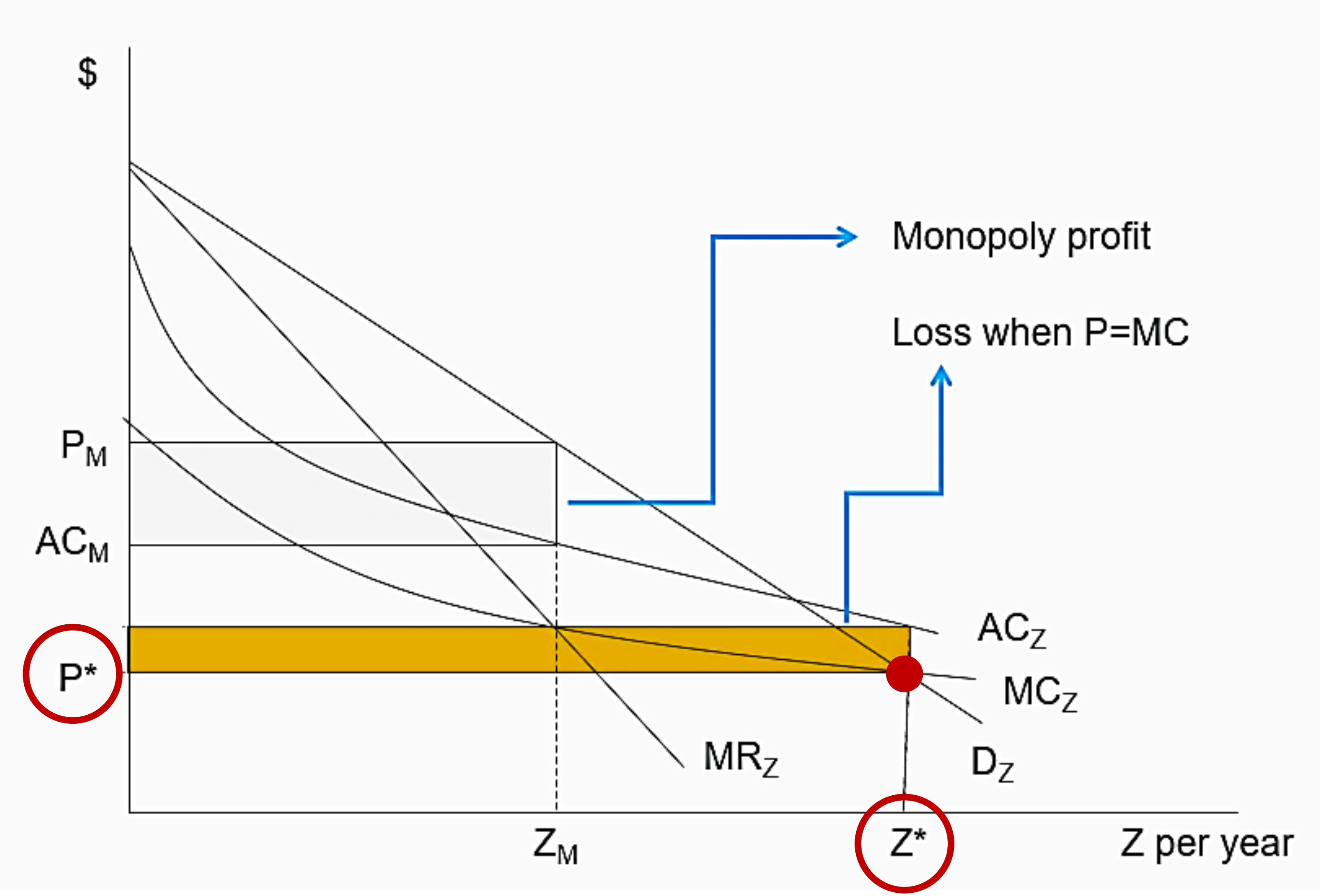
作業四Rosen 16.5

(Rosen 16.5) Suppose that a town is considering a project to install new underground water pipes. Some of the costs are fixed in the sense that they do not increase with an increase in the amount of water consumed. For example, pipes deteriorate over time, independent of the volume of water flowing through them. Hence, it is impossible to pay for the investment in the pipes by charging consumers an amount based on the marginal cost of the water they consume. Under these circumstances, what might an efficient pricing system look like?

作業四Rosen 16.5

Efficiency generally requires setting price equal to MC, but doing so here would result in a price lower than the average cost, which would be unsustainable（因為虧損）. In this case, the problem of paying for fixed costs could be solved by charging a price equal to MC and levying a lump sum tax to pay for the fixed costs associated with the pipes.

作業四Rosen 16.5



邊際成本訂價法達到效率，

可是廠商有虧損

二部式訂價法：

以邊際成本訂價收使用費；

並對使用者收取定額稅，

為基本費，

彌補邊際成本訂價法下的虧損

作業五Rosen 17.3

Singh, who has a federal personal income tax rate of 28 percent, holds an oil stock that appreciates in value by 10 percent each year. He bought the stock one year ago.

Singh's stockbroker now wants him to switch the oil stock for a gold stock that is equally risky. Singh has decided that if he hold the oil stock, he will keep it only one more year and then sell it. If he sells the oil stock now, he will invest all the (after-tax) proceeds of the sale in the gold stock and then sell the gold stock one year from now. What is the minimum rate of return the gold stock must pay for Singh to make the switch? Relate your answer to the lock-in effect.

對於實現的利益課稅會有閉鎖效果，使資產持有者不願交易資產

作業五Rosen 17.3

持有oil stock已經1年，有10%的獲利，如果實現要繳28%的稅

Singh有2個選擇：

1)不賣，持有到明年賣掉：獲利再明年一起繳稅

2)賣掉，轉買gold stock到明年賣掉：今年獲利了結課一次，明年又了結一次，課2次稅

因為轉買gold stock要課2次稅，持續持有oil stock只要課一次，如果gold stock報酬沒有高於一定水準，儘管gold stock報酬率更佳，還是會選擇持有oil stock→閉鎖效果

option 1 ：

1000*(1.1)^2 = 1210；獲利210要課28%，淨利=210*(1-0.28)=151.2，稅後本利和1151.2

option 2 ：

1000*1.1 = 1100(獲利了結)；淨利=100*(1-0.28)=72，稅後本利和1072

1072投入gold stock→1年後變成1072*(1+r)，

本利和=1072+1072*r*(1-0.28)，必須大於1151.2才願意投資，r=10.261194%

作業五Rosen 17.6

4. (Rosen 17.6) Suppose that a typical taxpayer has a marginal income tax rate of 35 percent. The nominal interest rate is 13 percent, and the expected inflation rate is 8 percent.

a. What is the real after-tax rate of interest?

b. Suppose that the expected inflation rate increases by 3 percentage points to 11 percent, and the nominal interest rate increases by the same amount. What happens to the real after-tax rate of return?

c. If the inflation rate increases as in part b, by how much would the nominal interest rate have to increase to keep the real after-tax interest rate at the same level as in part a? Can you generalize your answer using an algebraic formula?

作業五Rosen 17.6

(a)

13%*(1-0.35)-8%=0.45%

(b)

(13%+3%)*(1-0.35)-11%=-0.6%

(c)

(13%+X%)*(1-0.35)-11%=0.45%

X=4.6153846

$(1-t)i-\pi=(1-t)i'-\pi'$.

$(\pi'-\pi)=(1-t)(i'-i)$; $(i'-i)= (\pi'-\pi)/(1-t)$.

$\Delta i=\Delta\pi/(1-t)$